

Final Review

ECE341

Combinational logic

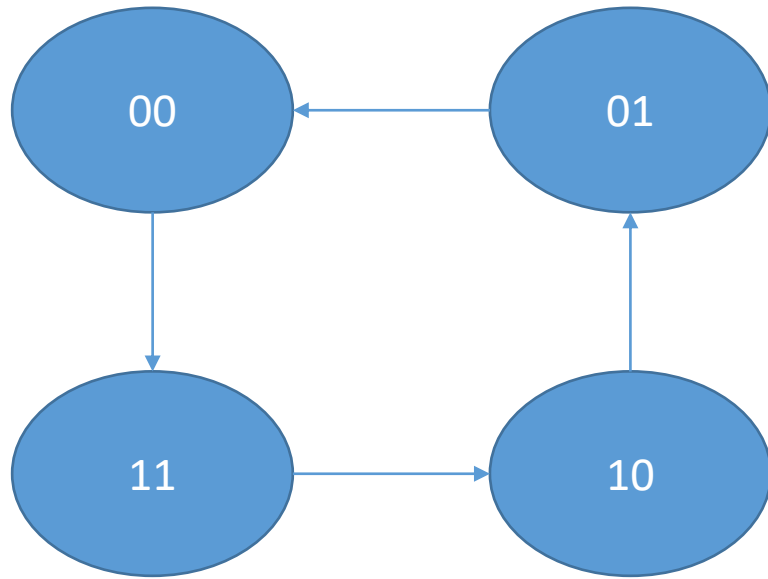
- Truth table
- K-map
- De-Morgan

Synchronous counter

- B1: Lập bảng mô tả trạng thái hiện tại, trạng thái tiếp theo của counter và giá trị ngõ vào của các FF tương ứng.
- B2: Tối thiểu hóa hàm bằng K-map cho các trạng thái ngõ vào của FF.
- B3: Vẽ mạch theo các hàm chức năng ngõ vào.

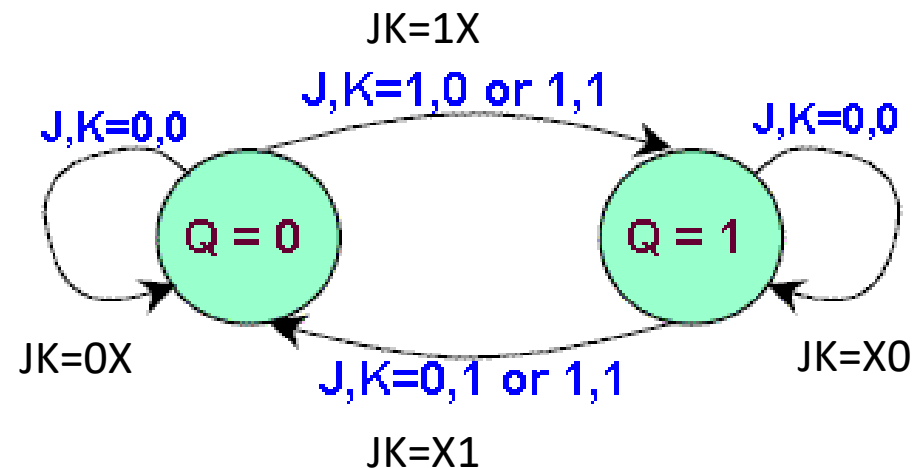
Synchronous counter

- Dùng JK FF thiết kế bộ đếm đồng bộ có các giá trị đếm



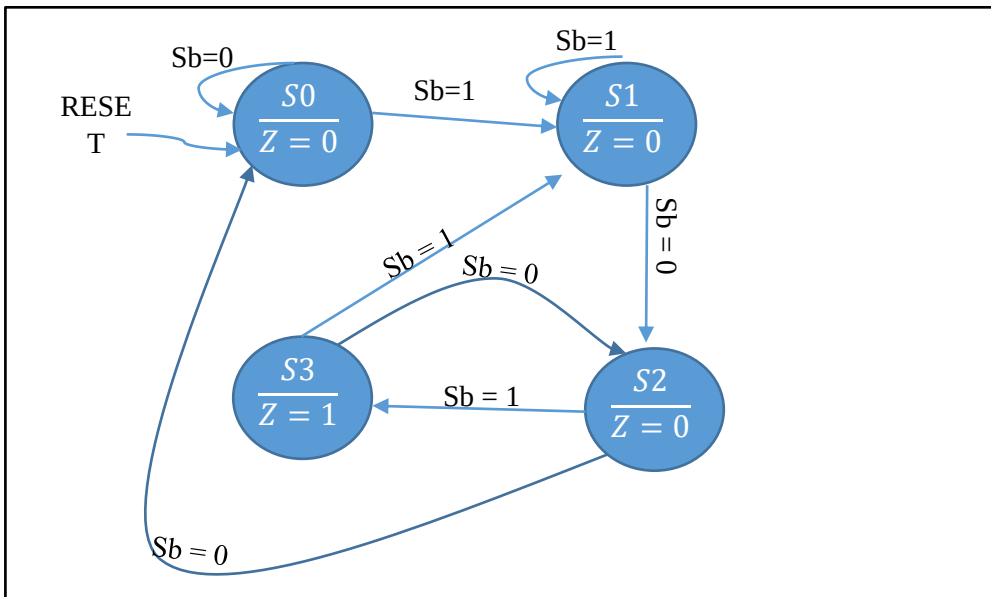
Trạng thái hiện tại [Q1Q0]	Trạng thái kế tiếp [Q1Q0]	Ngõ vào [J1K1]	Ngõ vào [J0K0]
00	11	1X	1X
01	00	00	X1
10	01	X1	1X
11	10	X0	X1

J	K	CLK	Q	$\bar{Q}$	Comment
0	0	$\uparrow$	Q	$\bar{Q}$	Latch
1	0	$\uparrow$	1	0	SET
0	1	$\uparrow$	0	1	RESET
1	1	$\uparrow$	$\bar{Q}$	Q	TOGGLE
X	X	Any Thing Else	Q	$\bar{Q}$	NO Change !



FSM - Finite State Machine design

Design Moore FSM detect pattern of serial data input, output $z=1$ if pattern is 101, including overlap.



CurrentState	NextState		Output Z
	Sb=0	Sb=1	
S0	S0	S1	0
S1	S2	S1	0
S2	S0	S3	0
S3	S2	S1	1

FSM

State	Q1Q0
S0	00
S1	01
S2	10
S3	11

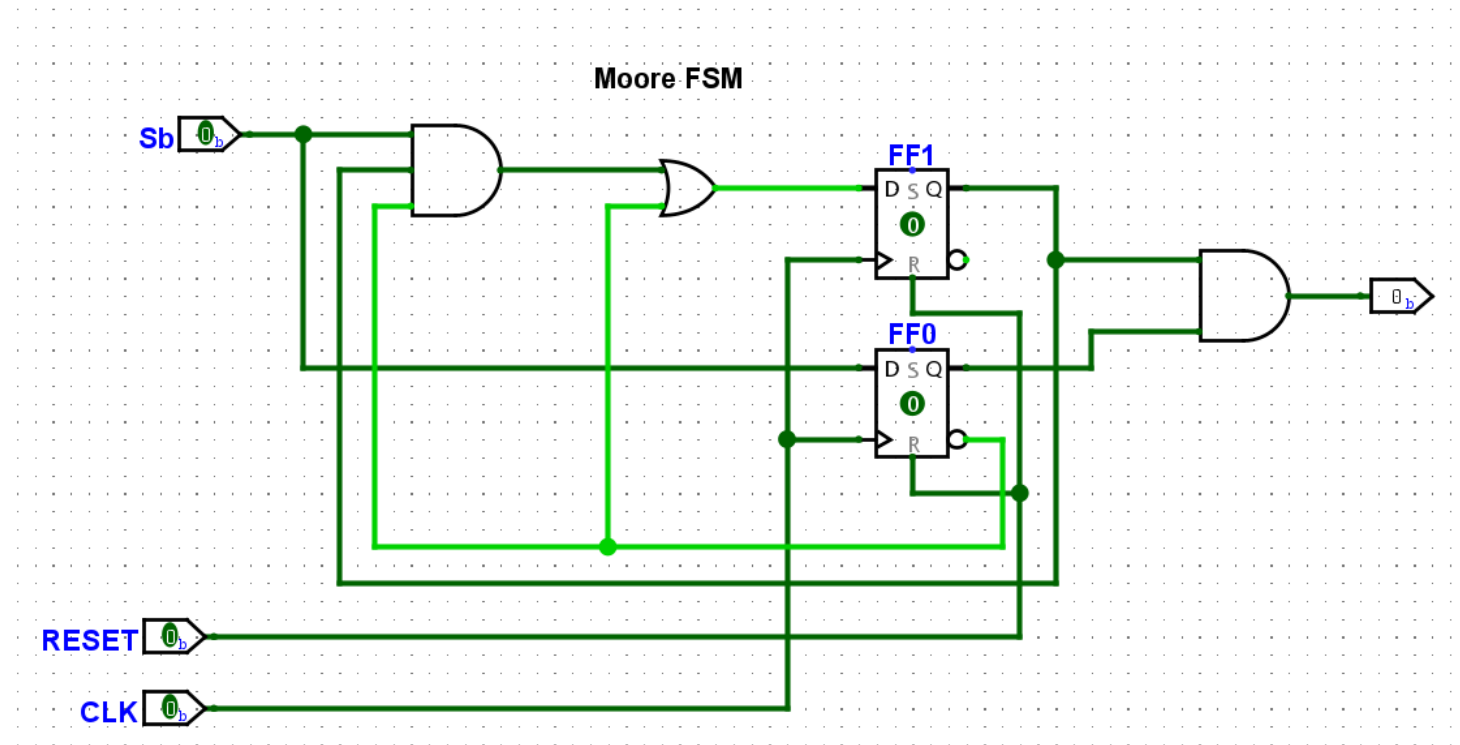
CurrentState	NextState		Output Z
	Sb=0	Sb=1	
S0	S0	S1	0
S1	S2	S1	0
S2	S0	S3	0
S3	S2	S1	1



Current state Q1Q0	NextState Q1Q0		Output Z
	Sb=0	Sb=1	
00	00	01	0
01	10	01	0
10	00	11	0
11	10	01	1

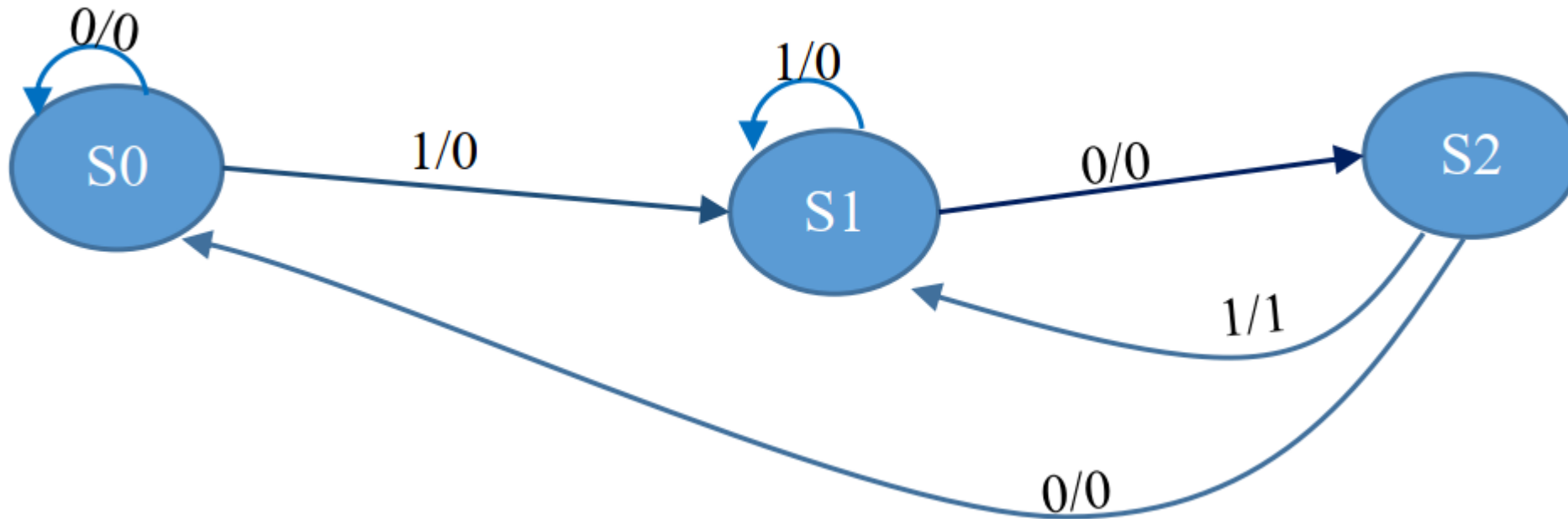
FSM

Q1Q0	Q1Q0 Sb=0	Q1Q0 Sb=1	D1D0 Sb=0	D1D0 Sb=1
00	00	01	00	01
01	10	01	10	01
10	00	11	00	11
11	10	01	10	01

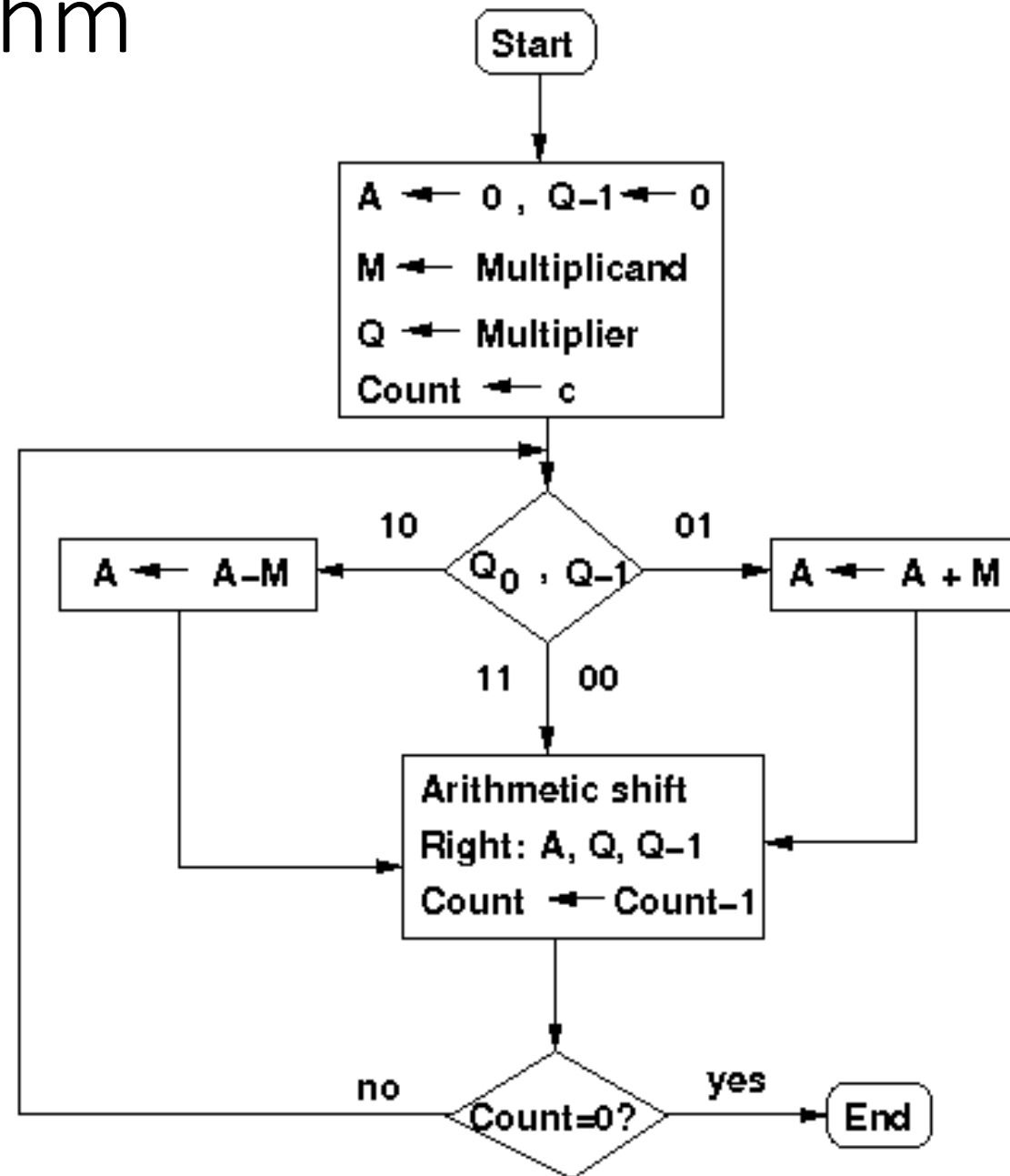


FSM

Design Mealy FSM detect pattern of serial data input, output $z=1$ if pattern is 101, including overlap.



Booth algorithm

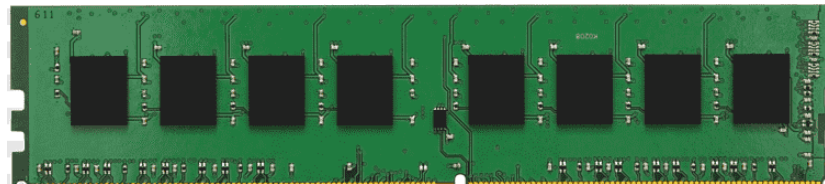


Booth algorithm

A	Q	Q-1	M		
0000	0101	0	0111	Initial value	
1001 1100	0101 1010	0 1	0111 0111	$A \leftarrow A - M$ shift	First cycle
0011 0001	1010 1101	1 0	0111 0111	$A \leftarrow A + M$ shift	Second cycle
1010 1101	1101 0110	0 1	0111 0111	$A \leftarrow A - M$ shift	Third cycle
0100 0010	0110 0011	1 0	0111 0111	$A \leftarrow A + M$ shift	Fourth cycle

Computer Memory

- Design a memory 64KB from 8KB memory chip. On each chip show the address range memory in hexadecimal.
- Design a memory 4KB from 2KB and 1KB memory chip. On each chip show the address range memory in hexadecimal.



Computer performance

$$T = \frac{N \times CPI}{R}$$

- T : Time required to execute a program
- N : Number of instructions executed to complete a program
- CPI : Average number of clock cycles per instruction
- R : Clock rate

Computer performance

Consider a program running on a processor P1 with a clock period of 0.5 nanoseconds. The instructions in the program can be categorized into two types:

Type-1: Each of these instructions takes 3 cycles to execute on P1.

Type-2: Each of these instructions takes 1 cycle to execute on P1.

Assume that $x\%$ of the total instructions in the program are of Type-1, while the remaining instructions are of Type-2. Also assume that processor is required to have an instruction execution rate of at least 1 billion instructions per second. What is the maximum value of “ x ” (percentage of Type-1 instructions), which will allow the processor to meet the desired execution rate target? Show all the necessary calculations.

Computer performance

$$\text{CPI} = (x * 3) + ((1 - x) * 1) = 2x + 1$$

$$\text{Clock Period} = 0.5 \text{ ns} = 0.5 * 10^{-9} \text{ s}$$

$$\text{Clock Rate} = 1 / \text{Clock Period} = 2 \text{ GHz}$$

$$\text{Required Instruction Execution Rate} = 10^9 \text{ instructions per second}$$

$$\text{Since: Execution time} = (\text{Number of Instructions} * \text{CPI}) / \text{Clock Rate}$$

$$\text{Therefore: Instruction Execution Rate} = \text{Number of Instructions} / \text{Execution Time} = \text{Clock Rate} / \text{CPI}$$

In order to meet the desired execution rate target:

$$\text{Clock Rate} / \text{CPI} \geq 10^9$$

$$2 * 10^9 / (2x + 1) \geq 10^9$$

$$\text{Solving for } x \text{ yields: } x \leq 0.5$$

Therefore the percentage of Type-1 instructions in the program must be less than 50%.

Computer performance

- A company is designing a processor P1 to execute a program, whose instruction statistics are as follows:

Type of instruction	Percent of total instructions	Cycle per instructions
Branch	10%	3
Load	30%	5
Store	20%	4
Arithmetic	40%	4

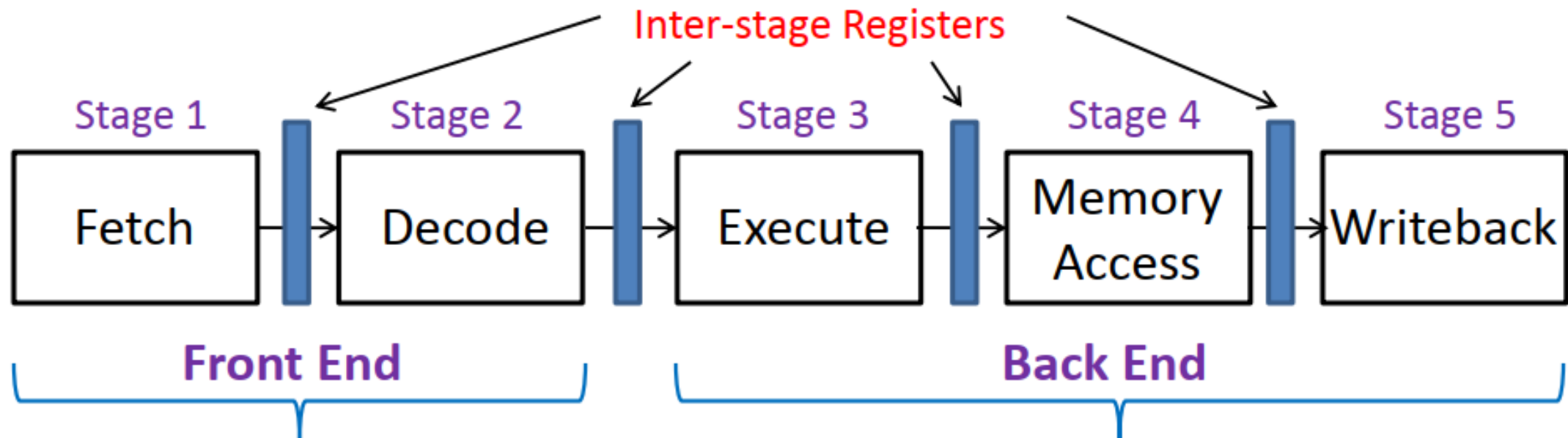
- The chief architect for the processor sets a target that the processor must be able to execute 500 million instructions per second. What is the minimum clock speed that the processor must run at to achieve this target?

Computer performance

- $\text{CPI} = (10\% * 3) + (30\% * 5) + (20\% * 4) + (40\% * 4) = 4.2$
- Instruction Execution Rate = 500 million instructions per second = $5 * 10^8$ instructions per second
- Since:
Execution time = (Number of Instructions * CPI) / Clock Rate
- Therefore:
Clock Rate = (Number of Instructions * CPI) / Execution Time = CPI * Instruction Execution Rate
 $= 4.2 * 5 * 10^8 = 2.1 * 10^9 = 2.1 \text{ GHz}$
- Therefore the processor must operate at a clock speed of at least 2.1 GHz to satisfy the instruction execution rate target.

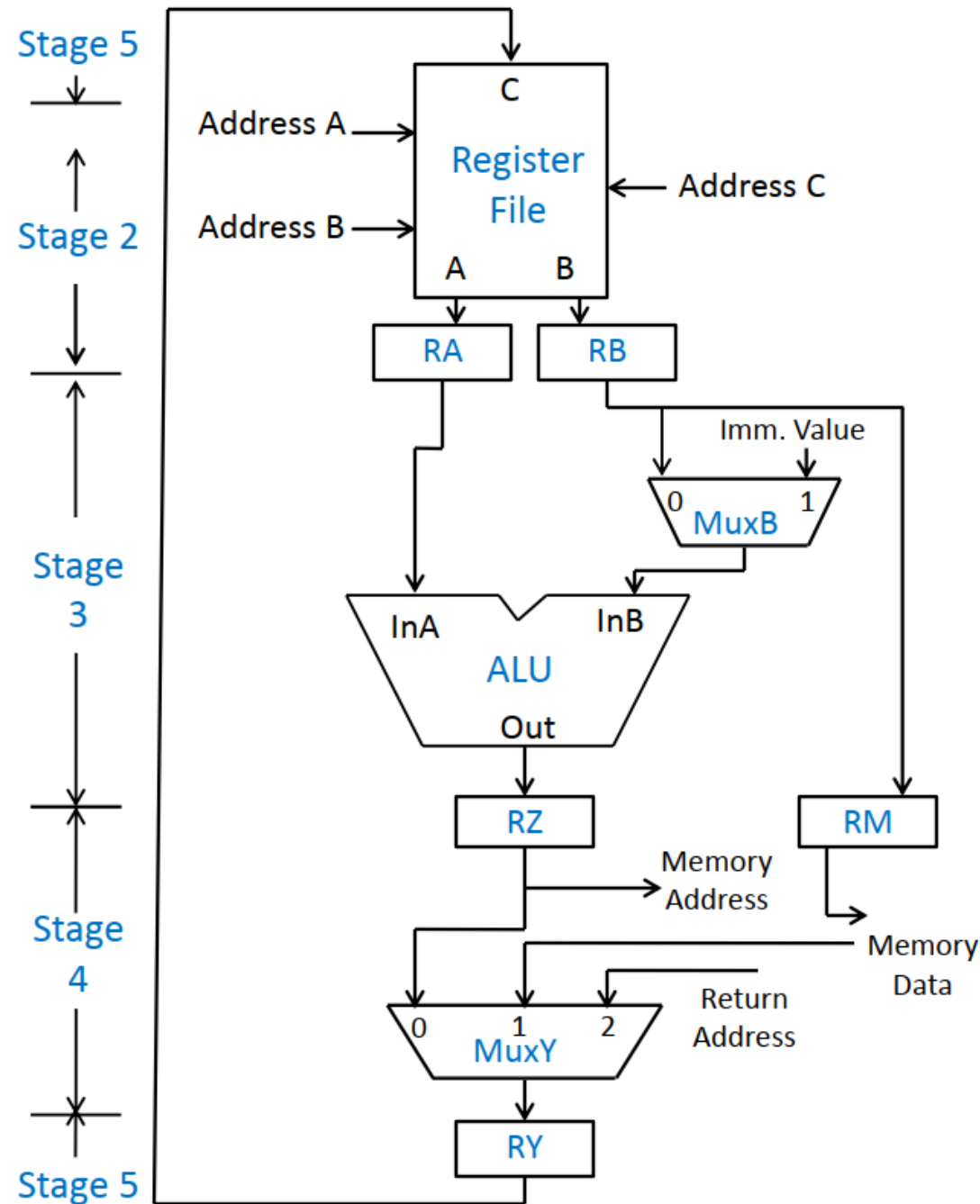
RISC Processor datapath

- **Datapath** is a combination of register read, back-end stages and inter-stage registers



RA, RB, RZ, RM and RY
are **inter-stage**
registers

Register file is in both
stage 2 and stage 5,
because stage 2 (Decode)
reads source registers and
stage 5 (Writeback) writes
into destination registers



Example 1: Add R3, R4, R5

Stage 1:

Memory address $\leftarrow$ [PC],
Read memory,
IR $\leftarrow$ Memory data,
PC $\leftarrow$ [PC] + 4

Stage 2:

Decode instruction,
RA $\leftarrow$ [R4], RB $\leftarrow$ [R5]

Stage 3:

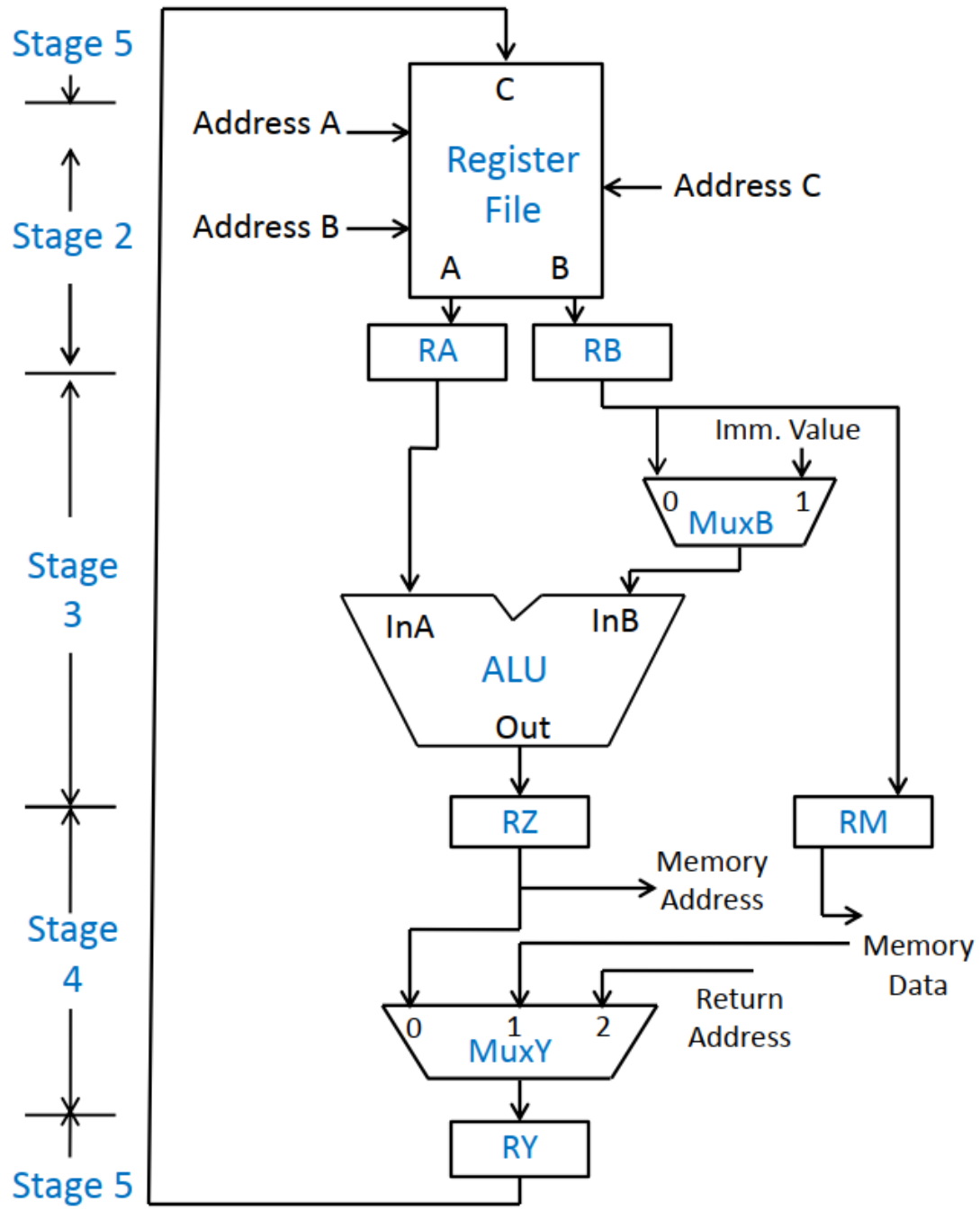
RZ $\leftarrow$ [RA] + [RB]

Stage 4:

RY $\leftarrow$ [RZ]

Stage 5:

R3 $\leftarrow$ [RY]



Example 2: Load R5, X(R7)

Stage 1:

Memory address $\leftarrow$ [PC],
Read memory,
IR $\leftarrow$ Memory data,
PC $\leftarrow$ [PC] + 4

Stage 2:

Decode instruction,
RA $\leftarrow$ [R7]

Stage 3:

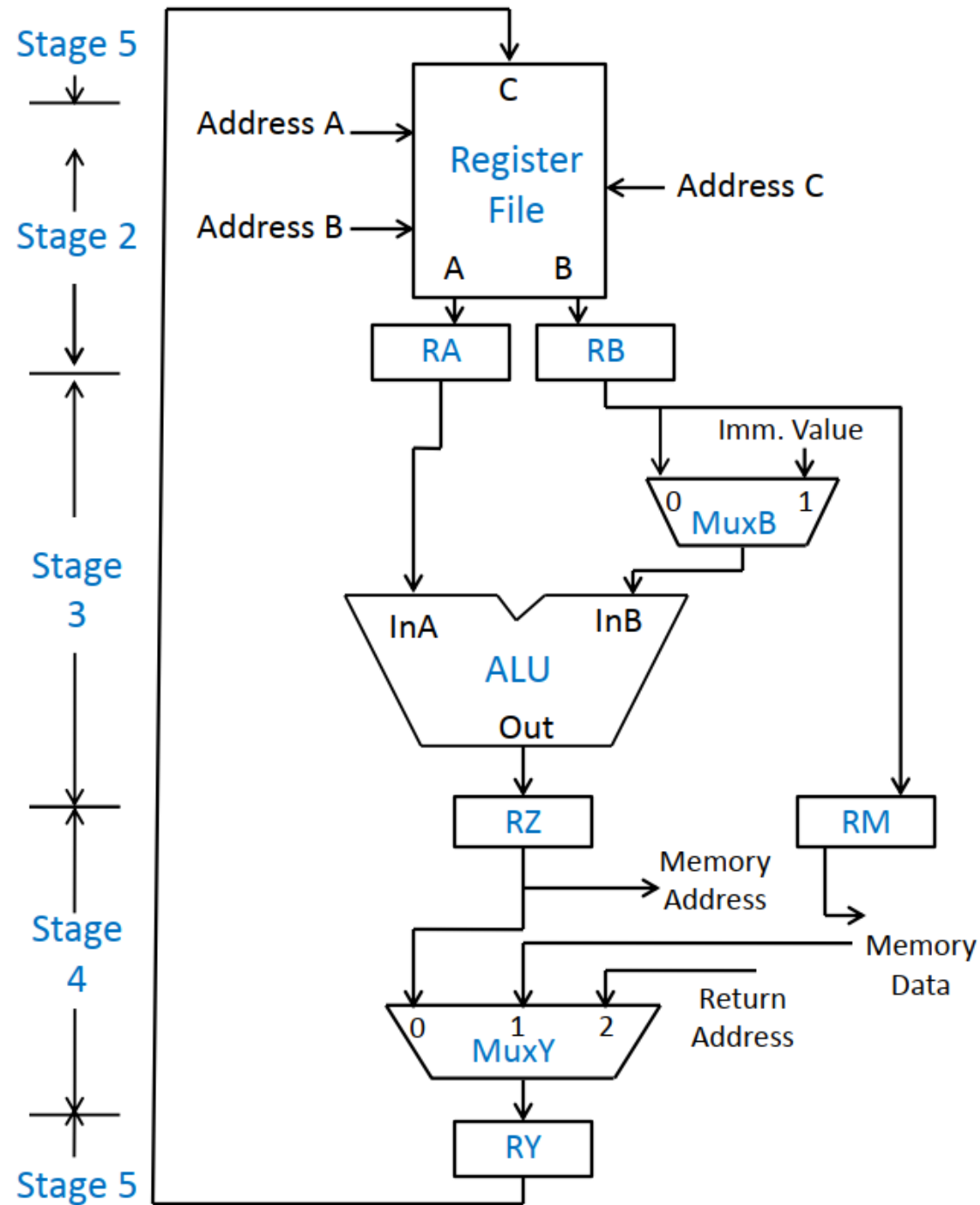
RZ $\leftarrow$ [RA] + Immediate value X

Stage 4:

Memory address $\leftarrow$ [RZ],
Read memory
RY $\leftarrow$ Memory data

Stage 5:

R5 $\leftarrow$ [RY]



Example 3: Store R6,X(R8)

Stage 1:

Memory address $\leftarrow$ [PC],
Read memory,
IR $\leftarrow$ Memory data,
PC $\leftarrow$ [PC] + 4

Stage 2:

Decode instruction,
RA $\leftarrow$ [R8], RB $\leftarrow$ [R6]

Stage 3:

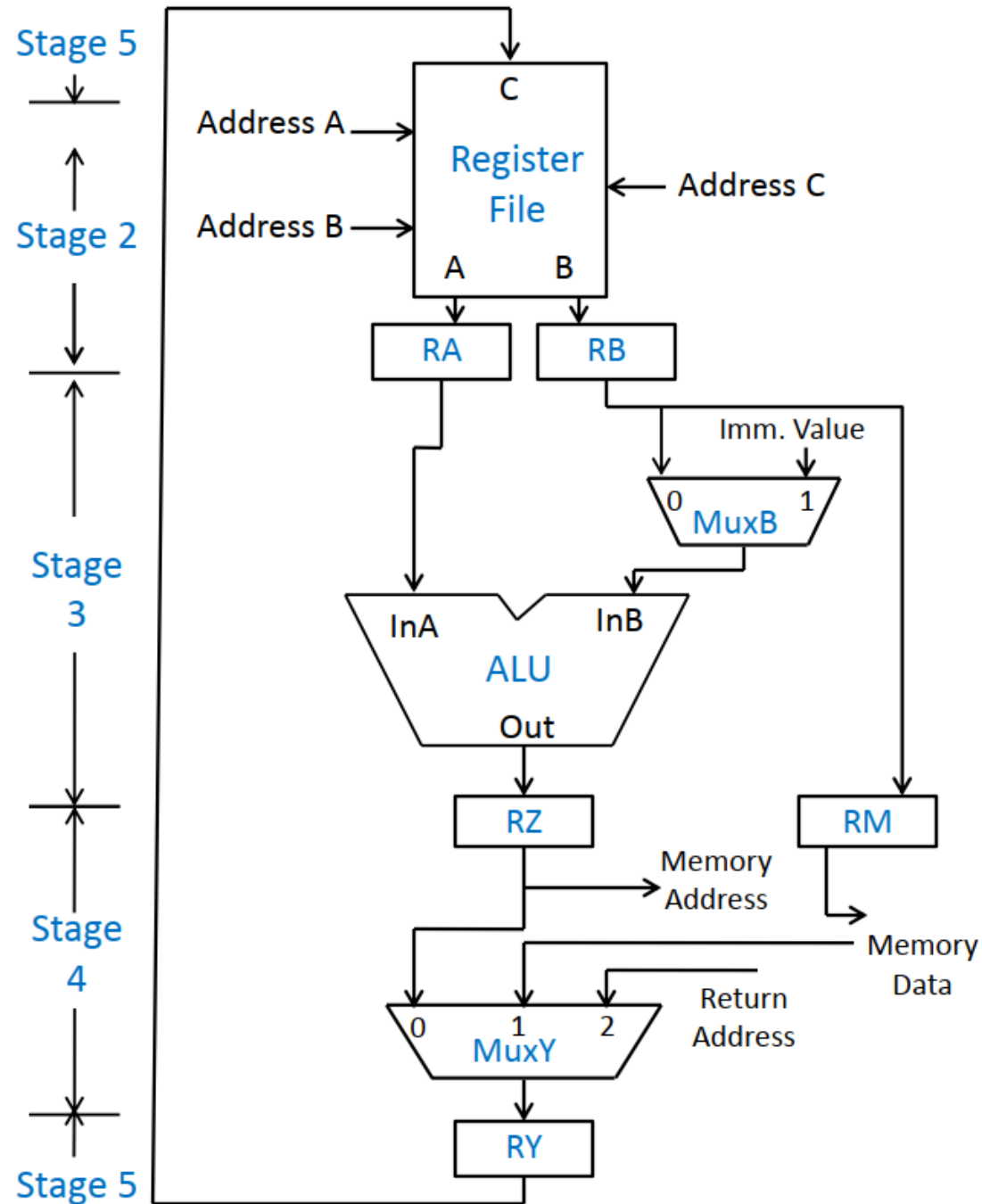
RZ $\leftarrow$ [RA] + Immediate value X
RM $\leftarrow$ [RB]

Stage 4:

Memory address $\leftarrow$ RZ,
Memory data $\leftarrow$ [RM],
Write memory

Stage 5:

No action



The 5-stage RISC processor discussed in class is used to execute the following

sequence of instructions, one after the other:

Instruction-1: LOAD R4, #200(R2)

Instruction-2: AND R5, R3, R4

Instruction-3: CALL_REGISTER R5

The processor data path with all the control signals is shown in the following figure:

Fill the following table to indicate the values of control signals for each of the three instructions:

Y-select in stage-4

RF_write in stage-5

C_select in stage-5

